

1. Argand Atrium — Complex Numbers
 - a. I definition

Example 1

Solve the equation in each case.

a) $x^2 = -25$

b) $4x^2 + 484 = 0$

c) $x^2 + 20 = 0$

a) $x^2 = -25$

$$x = \pm\sqrt{-25}$$

$$x = \pm 5i$$

Take **both** square roots of -25 .

Write $\sqrt{-25}$ as a multiple of i .

b) $4x^2 + 484 = 0$

$$x^2 = -121$$

$$x = \pm\sqrt{-121} = \pm 11i$$

Rearrange to find the value of x^2 .

c) $x^2 + 20 = 0$

$$x = \pm\sqrt{-20} = \pm\sqrt{20}i$$

$$= \pm 2\sqrt{5}i$$

Leave the value of x in exact terms.

You may leave your answer in surd form.

Complex numbers

Example 2

Simplify:

a) i^3

b) i^4

c) i^9

d) $(-i)^6$

e) $(2i)^5$

a) $i^3 = i^2 \times i = -1 \times i = -i$

Replace i^2 with -1 .

b) $i^4 = i^2 \times i^2 = -1 \times -1 = 1$

c) $i^9 = i^4 \times i^4 \times i = 1 \times 1 \times i = i$

Write i^9 as a product of powers of i each of which you can replace with 1 , -1 , i or $-i$. Use $i^4 = 1$ here.

d) $(-i)^6 = i^6 = i^4 \times i^2 = 1 \times -1 = -1$

Expand the brackets and simplify.

e) $(2i)^5 = 32 \times i^5 = 32 \times i^4 \times i = 32i$

It is useful to remember:
 $i^2 = -1$, $i^3 = -i$, $i^4 = 1$.

Use the ordinary laws of algebra to write the expression without brackets, then simplify.

b. Solving equations using complex numbers

Example 3

Solve the equation $2z^2 - z + 3 = 0$.

Using the quadratic formula,

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$z = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times 2 \times 3}}{2 \times 2}$$

$$z = \frac{1 \pm \sqrt{1 - 24}}{4}$$

$$z = \frac{1 \pm \sqrt{-23}}{4}$$

$$z = \frac{1}{4} \pm \frac{\sqrt{23}}{4}i$$

Substitute $a = 2$, $b = -1$ and $c = 3$ into the formula.

Simplify the expression.

Here we leave the answer in the form $a \pm bi$, but we could separate the two values and write

$$z = \frac{1}{4} + \frac{\sqrt{23}}{4}i \text{ and } \frac{1}{4} - \frac{\sqrt{23}}{4}i.$$

Alternatively, completing the square,

$$2z^2 - z + 3 = 0$$

$$2\left(z^2 - \frac{1}{2}z\right) + 3 = 0$$

$$2\left\{\left(z - \frac{1}{4}\right)^2 - \frac{1}{16}\right\} + 3 = 0$$

$$2\left(z - \frac{1}{4}\right)^2 - \frac{1}{8} + 3 = 0$$

$$2\left(z - \frac{1}{4}\right)^2 = -\frac{23}{8}$$

$$\left(z - \frac{1}{4}\right)^2 = -\frac{23}{16}$$

$$\left(z - \frac{1}{4}\right) = \pm\sqrt{-\frac{23}{16}}$$

$$z = \frac{1}{4} \pm \sqrt{-\frac{23}{16}}$$

$$z = \frac{1}{4} \pm \frac{\sqrt{23}}{4}i$$

Example 5

- a) Find the value of the real number p such that $p + (2 - 3i)(1 + 5i)$ is an imaginary number.
b) Find the values of the real numbers x and y such that $4(x + iy) = -2y - 3ix - 5(3 + 2i)$.

a) $p + (2 - 3i)(1 + 5i) = p + (2 - 3i + 10i - 15i^2)$

Expand the brackets and simplify.

$$= p + 7i + 17$$

$$= (p + 17) + 7i$$

Write in the form $x + iy$.

$$p + 17 = 0$$

$$p = -17$$

Equate the real part to 0 and solve to find p .

b) $4x + 4iy = -2y - 3ix - 15 - 10i$

$$4x + 4iy = (-2y - 15) + (-3ix - 10i)$$

Collect the real parts and the imaginary parts on the RHS.

$$\text{Comparing real parts, } 4x = -2y - 15$$

$$\text{Comparing imaginary parts, } 4y = -3x - 10$$

$$4x + 2y = -15 \quad (1)$$

$$3x + 4y = -10 \quad (2)$$

Solve the simultaneous equations.

$$2 \times \text{equation (1)} - \text{equation (2):}$$

$$5x = -20$$

$$x = -4$$

$$\text{Substituting } x = -4 \text{ into equation (1):}$$

$$4 \times -4 + 2y = -15$$

$$y = 0.5$$

$$x = -4, y = 0.5$$

Example 6

The complex number $3 + 2i$ is a root of the quadratic equation $z^2 - (5 + 2i)z + a + bi = 0$, where a and b are real.

- a) Find the values of a and b .
b) **Explain** why the two roots are not complex conjugates of each other.

- a) As $z = 3 + 2i$ is a root,

$$(3 + 2i)^2 - (5 + 2i)(3 + 2i) + a + bi = 0$$

$$(9 + 12i + 4i^2) - (15 + 16i + 4i^2) + a + bi = 0$$

$$-6 - 4i + a + bi = 0$$

$$-6 + a + (-4 + b)i = 0$$

$$-6 + a = 0 \text{ and } -4 + b = 0$$

$$a = 6, b = 4$$

Substitute for z in the equation.

Expand and then simplify.

Compare the real part on the LHS with the real part on the RHS, and similarly compare the imaginary parts.

- b) The coefficients of the quadratic equation $z^2 - (5 + 2i)z + a + bi = 0$ are not all real so the roots will not be complex conjugates.

Example 7

Find the square roots of the complex number $7 - 24i$.

As with any other number we expect to get two square roots of any complex number.

To find the square roots, solve the equation

$z^2 = 7 - 24i$, where $z = x + iy$, where x and y are real.

$$(x + iy)^2 = 7 - 24i$$

$$x^2 + 2xyi + i^2y^2 = 7 - 24i$$

$$x^2 + 2xyi - y^2 = 7 - 24i$$

$$x^2 - y^2 = 7 \quad (1)$$

$$2xy = -24 \quad (2)$$

Substitute $x + iy$ for z .

Expand the bracket and simplify.

Compare real and imaginary parts to get simultaneous equations.

From (2) $y = \frac{-12}{x}$ and substitute for y in equation (1):

$$x^2 - \left\{ \frac{-12}{x} \right\}^2 = 7$$

$$x^2 - \frac{144}{x^2} = 7$$

$$x^4 - 144 = 7x^2$$

$$x^4 - 7x^2 - 144 = 0$$

$$(x^2 - 16)(x^2 + 9) = 0$$

$x^2 = 16$ or $x^2 = -9$, but x is real so $x = \pm 4$ only.

Substitute $x = 4$ and $x = -4$ into (2):

$$y = \frac{-12}{\pm 4} \Rightarrow y = \mp 3$$

$$x = 4, y = -3 \text{ or } x = -4, y = 3$$

The square roots of $7 - 24i$ are $4 - 3i$ and $-4 + 3i$.

Eliminate one variable to get a quartic equation in x .

Solve the equation to get the possible values of x .

Find the corresponding y values.

Note that these square roots may be written in the form $\pm(4 - 3i)$.

Example 8

Given that $2 + i$ is a root of the equation $z^3 - z^2 - 7z + 15 = 0$, find the other two roots.

Since one complex root is $2 + i$ and the coefficients of the equation are all real, the conjugate $2 - i$ is also a root.

$[z - (2 + i)][z - (2 - i)]$ is a factor.

$$z^2 - (2 + i)z - (2 - i)z + (2 + i)(2 - i)$$

$$z^2 - 4z + (4 + 2i - 2i - i^2)$$

$$z^2 - 4z + 5 \text{ is a factor.}$$

To find the third root (which must be real),

$$z^3 - z^2 - 7z + 15 = (\text{linear factor}) \times (z^2 - 4z + 5)$$

By considering the term in z^3 and the constant term, we can see that the linear factor must be $(z + 3)$.

The other two roots are -3 and $2 - i$.

Write down the product of the known factors.

Multiply out and simplify.

In this case, we can find the other factor by inspection. An alternative method is to carry out a long division of $(z^3 - z^2 - 7z + 15)$ by $(z^2 - 4z + 5)$.

c. Complex numbers represented on argand diagram

Example 9

$$u = 2 - 3i$$

On a sketch of an Argand diagram, show the points representing the complex numbers u , u^* , u^2 , $u + u^*$, $u - u^*$, uu^* and $\frac{u}{u^*}$.

$$u^* = 2 + 3i$$

$$u^2 = (2 - 3i)(2 - 3i) = -5 - 12i$$

$$u + u^* = (2 - 3i) + (2 + 3i) = 4$$

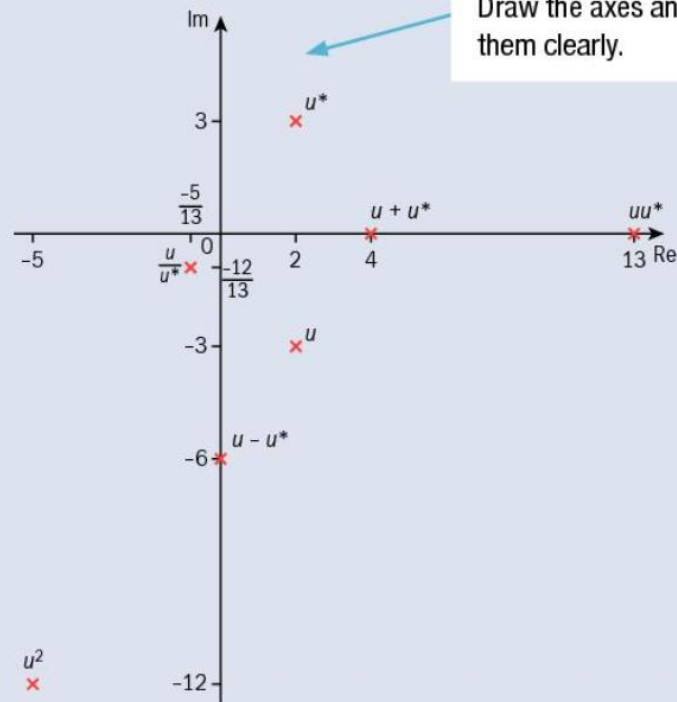
$$u - u^* = (2 - 3i) - (2 + 3i) = -6i$$

$$uu^* = (2 - 3i)(2 + 3i) = 13$$

$$\frac{u}{u^*} = \frac{2-3i}{2+3i} \times \frac{2-3i}{2-3i} = \frac{-5-12i}{13} = -\frac{5}{13} - \frac{12}{13}i$$

Calculate the complex numbers listed.

Draw the axes and plot your answers. Label them clearly.



d. Polar and exponential form of complex number

Example 10

- a) The complex numbers z_1 and z_2 are defined by $z_1 = 1 + i$ and $z_2 = -2 - 4i$.

Find the modulus and argument of z_1 and z_2 .

- b) The complex numbers z_3 and z_4 are defined by $z_3 = \left(5, \frac{\pi}{3}\right)$ and $z_4 = \left(2, \frac{5\pi}{6}\right)$.

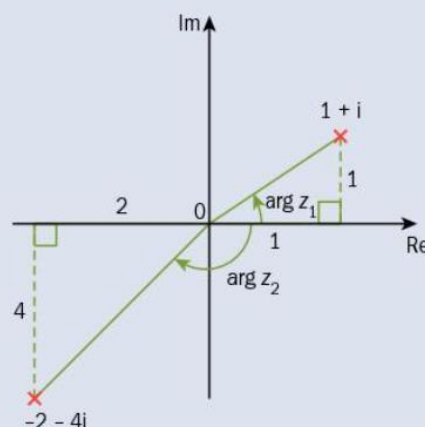
Write z_3 and z_4 in the form $x + iy$, where x and y are real numbers.

a) $|z_1| = \sqrt{1^2 + 1^2} = \sqrt{2} = 1.41 \text{ (3 s.f.)}$

$$\arg(z_1) = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4} = 0.785 \text{ (3 s.f.)}$$

$$|z_2| = \sqrt{(2)^2 + (4)^2} = \sqrt{20} = 4.47 \text{ (3 s.f.)}$$

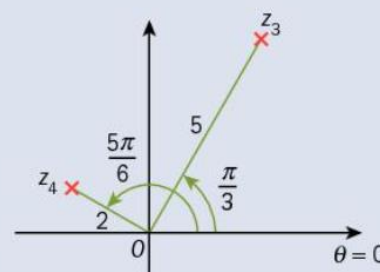
$$\arg(z_2) = -\left(\pi - \tan^{-1}\left(\frac{4}{2}\right)\right) = -2.03 \text{ (3 s.f.)}$$



It can be useful to sketch a diagram.

b) $z_3 = 5\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right) = \frac{5}{2} + \frac{5\sqrt{3}}{2}i = 2.5 + 4.33i$

$$z_4 = 2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right) = -\sqrt{3} + i = -1.73 + i$$



Example 11

The complex numbers z and w are defined by $z = 2 + 2i$ and $w = -1 + \sqrt{3}i$.

Find the modulus and argument of

- a) z^2 b) z^3 c) z^2w^2 d) $\frac{w^2}{z^3}$.

a) $z = (2\sqrt{2}, \frac{\pi}{4})$

Write z in modulus–argument form.

$$z^2 = \left((2\sqrt{2})^2, 2 \times \frac{\pi}{4} \right) = \left(8, \frac{\pi}{2} \right)$$

Use $(r, \theta)^2 = (r^2, 2\theta)$ and simplify.

$$|z^2| = 8, \arg(z^2) = \frac{\pi}{2}$$

b) $z^3 = \left((2\sqrt{2})^3, 3 \times \frac{\pi}{4} \right) = \left(16\sqrt{2}, \frac{3\pi}{4} \right)$

Use $(r, \theta)^3 = (r^3, 3\theta)$ and simplify.

$$|z^3| = 16\sqrt{2}, \arg(z^2) = \frac{3\pi}{4}$$

Alternatively, exponential form notation may be used.

For example, $z = 2\sqrt{2}e^{\frac{i\pi}{4}}$

$$\begin{aligned} z^2 &= \left(2\sqrt{2}e^{\frac{i\pi}{4}} \right)^2 \\ &= 8e^{\frac{i\pi}{2}} \text{ etc.} \end{aligned}$$

c) $w = \left(2, \frac{2\pi}{3} \right)$

$$w^2 = \left(2^2, 2 \times \frac{2\pi}{3} \right) = \left(4, \frac{4\pi}{3} \right)$$

Find w^2 and use your answer for z^2 from part (a).

$$z^2w^2 = \left(8, \frac{\pi}{2} \right) \times \left(4, \frac{4\pi}{3} \right) = \left(8 \times 4, \frac{\pi}{2} + \frac{4\pi}{3} \right)$$

Use $(r_1, \theta_1) \times (r_2, \theta_2) = (r_1 r_2, \theta_1 + \theta_2)$.

$$|z^2w^2| = 32, \arg(z^2w^2) = \frac{11\pi}{6} = -\frac{\pi}{6}$$

Write $\arg(z)$ so it lies between $-\pi$ and π .

d) $\frac{w^2}{z^3} = \left\{ \frac{4}{16\sqrt{2}}, \frac{4\pi}{3} - \frac{3\pi}{4} \right\} = \left\{ \frac{\sqrt{2}}{8}, \frac{7\pi}{12} \right\}$

Use $\frac{(r_1, \theta_1)}{(r_2, \theta_2)} = \left(\frac{r_1}{r_2}, \theta_1 - \theta_2 \right)$ and simplify.

$$\left| \frac{w^2}{z^3} \right| = \frac{\sqrt{2}}{8}, \arg\left(\frac{w^2}{z^3} \right) = \frac{7\pi}{12}$$

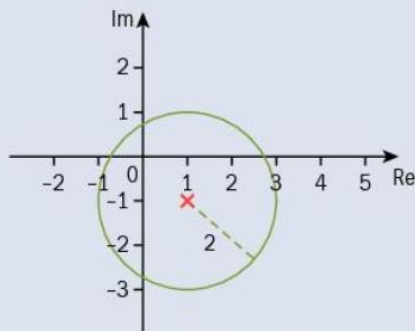
e. Loci in argand diagram

Example 12

- a) Sketch an Argand diagram and show all the points which represent the complex numbers z that satisfy the equation $|z - 1 + i| = 2$.
- b) The points in an Argand diagram representing $2 + 5i$ and $-6 + i$ are the ends of a diameter of a circle. Find the equation of the circle, giving your answer in the form $|z - (a + bi)| = k$.

a) $|z - (1 - i)| = 2$

z lies on a circle, centre $1 - i$ and radius 2.



Write the equation in the form $|z - z_1| = r$.

- b) The centre of the circle is given by

$$\frac{(2 + -6)}{2} + \frac{(5i + i)}{2} = -2 + 3i$$

The radius of the circle is given by

$$\frac{1}{2} \sqrt{(-6 - 2)^2 + (1 - 5)^2} = \frac{1}{2} \sqrt{80} = 2\sqrt{5}$$

The equation of the circle is $|z - (-2 + 3i)| = 2\sqrt{5}$.

Use the result that the mid-point of the line joining (x_1, y_1) and (x_2, y_2) is given by $\left\{ \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right\}$ and that the length of the line joining (x_1, y_1) and (x_2, y_2) is given by $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Example 13

Sketch an Argand diagram and shade the region whose points represent the complex numbers z which satisfy both the inequality $|z - (2 + 3i)| \leq 2$ and the inequality $|z - 4| > 3$.

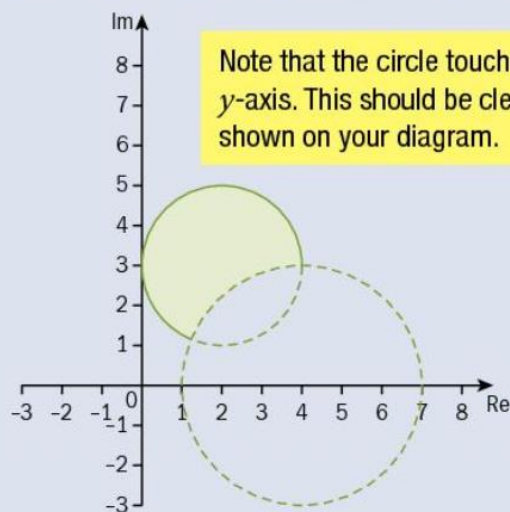
$|z - (2 + 3i)| \leq 2$

z lies inside or on the circumference of a circle, centre $2 + 3i$ and radius 2.

$|z - 4| > 3$

z lies outside a circle, centre $4 + 0i$ and radius 3.

The set of points which satisfies both conditions lies in the shaded region and on the part of the circle, centre $2 + 3i$, which is shown with a solid line.



Note that the circle touches the y -axis. This should be clearly shown on your diagram.

Example 14

- a) Sketch on an Argand diagram the loci given by $|z - (1 + 2i)| = 5$ and $|z - 5 + i| = |z + 3 - 5i|$.
 b) Show that these loci intersect at the point $-2 - 2i$.

a) $|z - (1 + 2i)| = 5$

z lies on a circle, centre $1 + 2i$
 and radius 5.

$$|z - 5 + i| = |z + 3 - 5i|$$

$$|z - (5 - i)| = |z - (-3 + 5i)|$$

z lies on the perpendicular bisector of
 the line joining $5 - i$ to $-3 + 5i$.

- b) The loci will intersect at $-2 - 2i$ if $z = -2 - 2i$
 satisfies both equations.

When $z = -2 - 2i$,

$$\begin{aligned} |z - (1 + 2i)| &= |-2 - 2i - (1 + 2i)| \\ &= |-3 - 4i| \\ &= 5 \end{aligned}$$

so $|z - (1 + 2i)| = 5$ is satisfied.

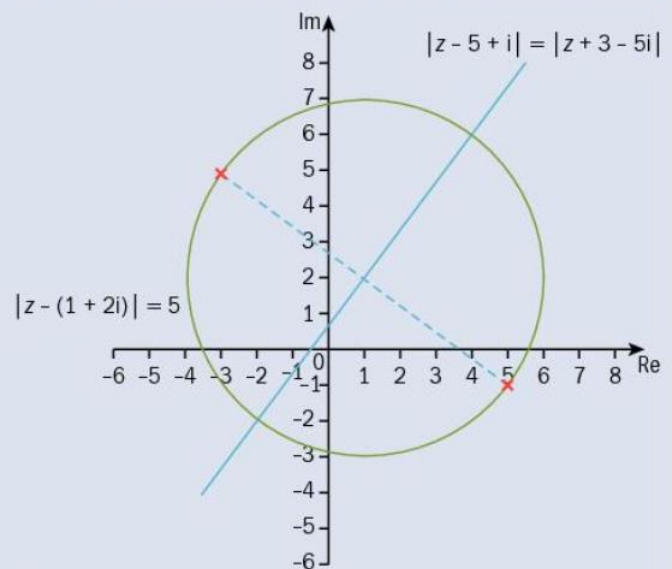
When $z = -2 - 2i$,

$$\begin{aligned} |z - 5 + i| &= |-2 - 2i - 5 + i| \\ &= |-7 - i| \\ &= \sqrt{50} \end{aligned}$$

$$\begin{aligned} \text{Also } |z + 3 - 5i| &= |-2 - 2i + 3 - 5i| = |1 - 7i| \\ &= \sqrt{50} \end{aligned}$$

so $|z - 5 + i| = |z + 3 - 5i|$ is satisfied.

Both conditions are satisfied and so these
 loci intersect at the point $-2 - 2i$.



Substitute $z = -2 - 2i$ into the left-hand side of the equation.

Substitute $z = -2 - 2i$ into each side of the equation in turn.

Make a clear statement in conclusion.

Example 15

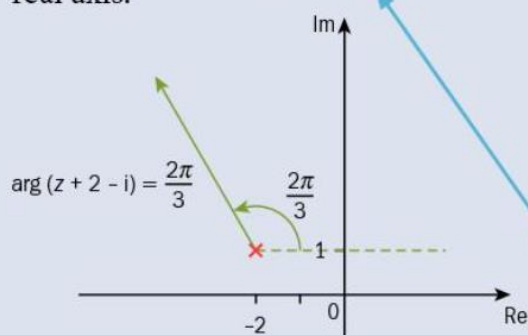
- a) Sketch on an Argand diagram the locus of the complex number z where z satisfies the equation $\arg(z + 2 - i) = \frac{2\pi}{3}$.
- b) On the same diagram, shade the region whose points represent the complex number z which satisfies $\frac{\pi}{2} \leq \arg(z + 2 - i) \leq \frac{2\pi}{3}$.

a) $\arg\{z - (-2 + i)\} = \frac{2\pi}{3}$

Write in the form $\arg(z - z_1) = \theta$.

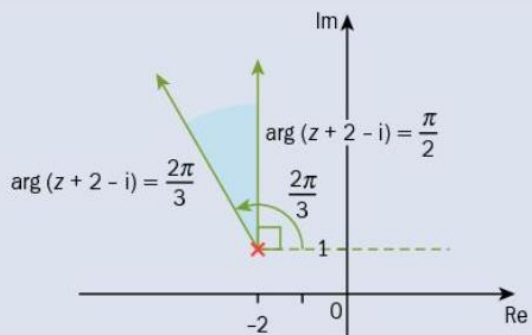
z lies on a half-line, starting at $-2 + i$, making an angle $\frac{2\pi}{3}$ with the positive real axis.

Identify z_1 and θ .



Draw a clear diagram. It is helpful to write a description to accompany your diagram.

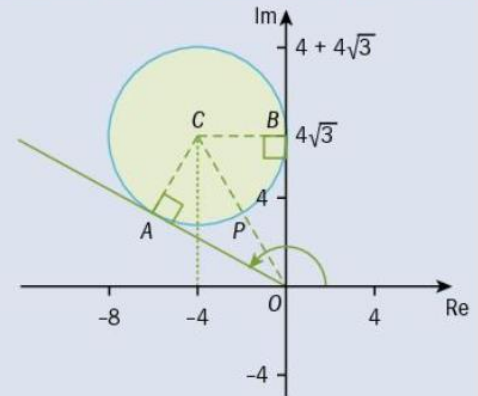
- b) z may lie anywhere in the shaded region or on its boundaries.



Example 16

- a) On a sketch of an Argand diagram, shade the region whose points represent the complex numbers z which satisfy the inequality $|z + 4 - 4\sqrt{3}i| \leq 4$.
- b) i) Find the least value of $|z|$ in this region.
 ii) Find the greatest value of $\arg z$ in this region.

- a) $|z - (-4 + 4\sqrt{3}i)| \leq 4$
 z lies on or inside a circle,
 centre $(-4 + 4\sqrt{3}i)$ and radius 4.



- b) i) The least value of $|z|$ is given by the length OP on the diagram.
 $OP = OC - PC$

$$\text{But } OC^2 = 4^2 + (4\sqrt{3})^2 = 16 + 48 = 64$$

Use Pythagoras' theorem on triangle OBC .

$$\text{So } OP = OC - PC = 8 - 4 = 4$$

Least value of $|z|$ is 4.

Use $PC = 4$ as it is a radius of the circle.

- ii) The greatest value of $\arg z$ is given by the angle OA makes with the positive real axis.

$$\text{Angle } COA = \sin^{-1}\left(\frac{CA}{OC}\right) = \sin^{-1}\left(\frac{4}{8}\right) = \frac{\pi}{6}$$

$$\text{The greatest value of } \arg z \text{ is } \frac{\pi}{6} + \frac{\pi}{6} + \frac{\pi}{2} = \frac{5\pi}{6}$$

$$\text{Angle } COA = \text{angle } COB = \frac{\pi}{6}$$